

Export Quality and Patent Protection: Stage-Dependent Effects in Development

Yan Liu¹, Walter G. Park^{2,*}, and Dahai Fu³

¹International Business School, Dalian Minzu University

²Department of Economics, American University

³School of International Trade and Economics,
Central University of Finance and Economics

September 2019

Abstract

This paper provides empirical evidence on the effects of patent protection on the vertical innovation of developing country exporters. We find that the relationship between export quality upgrading and patent protection is dependent on the technological stage of an economy. Specifically, the impact of patent protection depends on how close an economy's product quality is to the world frontier. We demonstrate that, for product qualities that are above some threshold closeness to this frontier, patent protection is positively associated with vertical innovation; for qualities below this threshold, patent protection is negatively associated with vertical innovation. We then characterize the density of the distribution of country products, determining which combinations fall under the threshold or pass the threshold, and draw some implications for patent policy design. To establish these findings, we employ a very extensive index of export product quality at the 4-digit SITC level and utilize a country-level measure of effective patent protection for the period 1990 - 2010 at five-year intervals.

Keywords: Patent Rights, Export Quality, Technological Stage Dependence, Developing Economies

JEL classification: O30, O34, F10, F14

***Corresponding author:** Walter G. Park, wgp@american.edu, 1-202-885-3774

1 Introduction

This paper investigates whether stronger patent protection induces exporters in developing countries to upgrade the quality of their products. Since the mid-1990s, the developing world has had to adopt stronger intellectual property (IP) provisions in order to meet the minimum standards set globally and to partake in the expanded global trade regime.¹ The strengthening of IP systems occurred despite the fact that developing economies were net importers of technologies, possessed relatively few and less valuable intellectual assets, and produced goods of comparatively lower product quality.

What motivates our investigation is two-fold. First, as discussed in Khandelwal (2010), upgrading export quality improves the chances of export success for developing countries competing in the new trading environment. Second, the premise is that stronger patent rights should spur innovation and lead eventually to an improvement in the product quality of developing country exports. Yet currently, evidence linking export quality to patent protection is absent. Recent work has shown that patent reforms can affect export capacity, but has not yet considered the effects on export quality (Maskus and Yang, 2018 and Ivus and Park, 2019). Studies exist on the relationship between innovative activity and export product quality (Akcigit et al., 2018 and Sampson, 2016), but this indicates only indirectly that patent rights can affect export quality via innovation. However, innovation responds to patent protection in diverse, complex ways – sometimes positively, other times adversely – that it would be useful to examine directly the relationship between export quality and patent rights.²

Our investigation shows why uniform (high) standards across countries is not suitable. The effects of patent protection on export quality vary by country and products. In particular, we estimate a threshold level of quality that is country and product-specific such that below this threshold, patent rights hinder vertical innovation; above it, they help promote quality upgrading. In this regard, we find the effects of patent protection on export quality to be *stage-dependent*;

¹Via agreements such as the *Trade-Related Aspects of Intellectual Property Rights (TRIPS) Agreement* among member-states of the World Trade Organization (WTO).

²See Saggi (2016) for a survey of the international IP literature.

that is, dependent on a country-product combination achieving a critical level of product quality. The intuition is that a sufficient level of technological capacity is required for bearing the increased costs of R&D development due to stricter patent protection and for generating sufficient market returns. Because of the stage-dependent effects, we argue that patent standards should ideally be differentiated by product and country.

In this paper, we use a quality ladders theoretical framework to guide our empirical analysis and discuss the Chinese ball-point pen industry as an example of how product quality and the IP regime varied by technological stage. For our empirical analysis, we exploit a rich dataset on detailed SITC 4-digit level export product qualities across forty-six developing countries over the period 1990 - 2010. Our empirical approach focuses on finding the critical threshold implied by the data. We then study how the threshold varies by income group. We find that the threshold is higher for upper middle income countries than it is for lower middle income countries because the products of upper middle income economies likely compete more intensely with the high quality products of advanced economies. For low income countries, we find no stage-dependent effects; patent protection only hinders their quality upgrading as their technological capacities are too low for IP incentives to have any consequence. We also study how the threshold varies by product group. We find the threshold to be higher for high IP and high patent intensity products since they require greater quality differentiation against existing IP-protected goods.

Lastly, we characterize the distribution of country-product qualities in the sample. We compare the density of country-product combinations above the threshold and below. Because the majority of combinations in a pooled sample are *above* the threshold, we deduce that on balance, stronger patent protection worldwide will increase export upgrading more than it will stifle it. However, we point out that the upgrading would disproportionately occur in the upper middle income countries; for lower middle income and low income countries, the quality upgrading of many country-products would be inhibited. The lesson again is that international IP policies should be attuned to gaps in technological development rather than be applied uniformly.

Our paper adds to existing work on export quality, which to our knowledge has not yet studied patent rights as a determinant.³ Recent work has focused on the effects of FDI⁴ and of trade liberalization.⁵ In this strand of the literature, Amiti and Khandelwal (2013) is the closest to our study in that they also analyze whether the effects of tariffs on export quality vary by closeness to the world frontier. A key difference is that we use a more recent measure of export quality derived from *world* exports data, whereas Amiti and Khandelwal (2013) use an index of quality based on exports to the U.S. only. Therefore, our paper develops a more generalized measure of the closeness to technological frontier, as we elaborate later.

Our paper also contributes to the recent literature emphasizing the stage-dependent effects of IPRs on innovation, economic growth, and development. Kim et al. (2012) find that countries at different stages of economic development utilize different types of IPRs, such as patents and utility models. Utility models are pervasive in the developing world since they are conducive to the kinds of incremental innovation that occur there. Chen and Puttitanun (2005) find that the optimal degree of IPR protection depends on the level of economic development. In less developed countries where indigenous innovative capacity is low, less stringent protection allows for useful catch-up imitation, while in countries where that capacity is sufficiently high, more stringent protection can stimulate innovative R&D. Hwang et al. (2016) provide a theoretical model to explain the U-shaped relationship between IPRs and economic development. For low-income countries the strength of IPRs falls with income since a low-income economy has not yet reached its critical market size for spurring innovation. Chu et al. (2014) find a stage-dependent effect of IPRs on innovation and economic growth using a Schumpeterian growth model of distance to frontier. The model shows that the growth-maximizing and welfare-maximizing levels of patent strength increase as a country evolves towards the world technology frontier.

The paper proceeds as follows: Section 2 provides a case study to motivate our analysis. Section

³See Schott (2004), Hallak (2006), Baldwin and Harrigan (2011), and Feenstra and Romalis (2014).

⁴See Harding and Javorcik (2012) and Anwar and Sun (2018).

⁵See Amiti and Khandelwal (2013) and Fan et al. (2015).

3 discusses our conceptual framework and section 4 the main data sources. Section 5 discusses our empirical framework and section 6 our results. Section 7 provides concluding thoughts.

2 Case study: the Ball-Point Pen Industry

The following industry case study helps illustrate a possible stage-dependent relationship between IPR protection and export quality.⁶ China is the largest exporter of ball-point pens in the world. In 2017, China exported 8.38 billion units to the rest of the world, accounting for 28.5% of world exports of such products, with Japan ranking second with a 14.3% share.⁷

The town of Fen Shui, nestled in Hang Zhou, Zhejiang province, is the manufacturing center of ball-point pens in China, accounting for around 40% of total national production. In 2009, the town was dubbed as the export hub of ball-point pens by the Ministry of Commerce of China. By 2017, 613 ball pen manufacturing companies and 379 supporting companies were in operation, making the town world renown for pen-making.

The pen business in Fen Shui started in the 1970s when China was largely an impoverished country. In the early stages, with limited resources and limited R&D capacity, the industry developed through imitation. Businesses expanded and competition was fierce and price-based. Firms that accumulated some technological capacity through decades of imitation attempted to beat the competition by undertaking some limited R&D. The aim was to climb the quality ladder by incorporating some minor attributes into the homogeneous ball-point pen. The minor quality differentiated ball-point pens provided a good return for the innovating company. However, in the absence of effective IPR protection, imitators inundated the industry and quickly turned the quality

⁶For this case study, we drew upon several materials. First, an interview with a sales manager of *Cixi Dengfeng Pen Manufacturing Co. Ltd.*, a leading company in the industry. Second, through company and association literature; in particular, by *Tonglu Guanghua Pen-making Co. Ltd.*, ranked first in sales in the Fen Shui region during 2014-2017 (<http://www.guanghuapen.com/>); *China Writing Instrument Association* (http://www.china-writing.com.cn/E_Show.aspx?type=82), and *China Light Industry Internet* (http://www.clii.com.cn/zhuantixinwen/jq/201707/t20170717_3910110.html).

⁷Export data can be found on www.Trademap.org using the class HS960810 for ball-point pens.

differentiated products back to homogeneous products. Minor product improvements developed by an innovator would diffuse throughout the whole industry at low cost. Thus, although intense imitative competition pushed prices down to the break-even point, the average quality for the industry tended to grow slightly over time as new attributes were built into the pens.

Disputes between innovators and imitators persisted as the size of the market and technological capacity of firms in the industry grew. As a solution to these disputes, a local administrative rule was enacted in the 1990s to maintain fair competition amid a weak national patent law system. Subsequently, to conform with the TRIPS agreement, China undertook a major reform of its patent system in 2000. The reforms continued, with China further amending its patent law by late 2009. As IPR protection had strengthened in China, innovators in Fen Shui pursued a strategy of quality differentiation and sought design patents. Independently, the quality of education in China improved and the knowledge base expanded from knowledge imported through foreign trade and inward FDI. These were fortuitous developments for innovators in Fen Shui as they were able to forge partnerships and cooperative ties with universities and industrial design centers. These ties were central to capacity-building and enabled firms to generate more design patents and upgrade their product qualities further, and with stronger IPRs, capture higher returns. In 2001, when China joined the WTO, few patent applications were filed by ball-point pen producers, but in 2002 the number of patent applications in Fen Shui surged to 110, and by 2006, it reached 541 applications. In 2017, over 4,100 patents were owned by ball-point pen producers in Fen Shui, accounting for 35% of national pen patents.

In brief, this case study provides contrasting experiences at different stages of technological development. In the early stages when the innovative capacity of pen producers was low, product quality growth occurred as the less stringent IPR regime enabled firms to focus on making small quality “jumps” and incorporating minor attributes into their product offerings. In the later stages, producers had acquired greater technological competencies through the support of complementary institutions and resources. Stronger IPRs – through either local administrative power or national

patent law enforcement – led pen producers to intensify their applications for patent rights. More rapid product quality growth followed. Reduced imitation risks raised producers’ expected profit margins and led them to focus on making larger quality “jumps.” The ball-point industry is but one example in which the impact of IPR protection on product quality depended on some threshold level of innovation potential. In the remainder of the paper, we present more formal evidence of stage dependence using a broad set of country-product data.

3 Conceptual Framework

We build on the themes in Chen and Puttitanun (2005), Chu et al. (2014), and Hwang et al. (2016) regarding the stage-dependent outcomes of IP protection. As a conceptual framework for guiding our empirical research, the quality ladders model of Grossman and Helpman (1991) augmented with an IPR factor is used.⁸ The purpose of what follows is to derive a model of vertical innovation and to show how it is a function of patent protection. We then motivate why the impact of patents on product quality will be stage-dependent.

On the demand side, consumers face the following intertemporal utility functional:

$$U = \int_0^{\infty} e^{-\rho t} \ln u(t) dt \quad (1)$$

where ρ is the time preference rate and $\ln u(t)$ the instantaneous utility function:

$$\ln u(t) = \int_0^1 \ln \sum_m q_{mj} x_{mj}(t) dj \quad (2)$$

where $x_{mj}(t)$ denotes the quantity of product j of quality m consumed at time t , and q_{mj} the corresponding quality level. Through innovations, product quality evolves as follows: $q_{mj} = \lambda(\kappa)q_{m-1j} = \lambda(\kappa)^m$, where $\lambda(\kappa)$ is the size of the quality jump and m the number of times the product climbed

⁸See Glass and Saggi (2002) for a more detailed quality ladders model incorporating IPRs.

up the quality ladder. The initial quality level is normalized to one (i.e., $q_{0j} = 1$). The size of the quality jump $\lambda(\kappa)$ is a function of the level of patent protection, given by κ . To the extent that stronger patent protection creates incentives for more radical innovations, $\lambda'(\kappa) > 0$. As will be shown below, a larger quality jump size enables firms to charge a higher markup (over a given amount of marginal cost) and earn a greater flow of instantaneous profits; this provides the incentive for investing in large step innovations rather than incremental ones.

Total expenditure at time t , $E(t)$, over a fixed measure of products is given by:

$$E(t) = \int_0^1 \ln p_{mj}(t) x_{mj}(t) dj \quad (3)$$

Static maximization of (2) subject to (3) yields the demand for product j of quality $\tilde{m}$ at time t : $x_{\tilde{m}j}(t) = \frac{E(t)}{p_{\tilde{m}j}(t)}$, where consumers purchase the quality level associated with the lowest quality-adjusted price, $\frac{p_{mj}(t)}{\lambda(\kappa)^m}$. Dynamic maximization of (1) subject to an intertemporal flow budget constraint $\dot{A}_t = rA_t + W_t - E_t$ (where A denotes assets, W income, and r the interest rate) and a transversality condition, yields the Euler equation $\frac{\dot{E}_t}{E_t} = r - \rho$. Assuming no international lending or borrowing, we get $r = \rho$ and constant expenditures, $E(t) = E$.

On the supply side, firms face the following value functional:

$$V = \int_0^\infty e^{-(\rho+\iota)t} \pi(t) dt \quad (4)$$

where ι (in the discount factor) captures the risk of losing this stream of profits due to an innovation of a new quality product that displaces the existing quality product.⁹ Thus, ι denotes innovation intensity. At this intensity, a firm that invests ιdt units of labor, over a time interval dt , has a probability of ιdt of successfully increasing product quality by one step, namely $\lambda(\kappa)$, and displacing a current incumbent and securing a new discounted stream of profits, V .

⁹In (4), there is symmetry among products such that $\pi_j(t) = \pi(t)$ because each leading brand is priced $\lambda(\kappa)$ times the marginal cost (namely the wage, w) in order to displace the previous leading brand.

We say a few more words about the parameter a . This parameter acts as a measure of the inverse productivity of R&D: the lower it is, the more efficient research is, as fewer resources (labor) are required to conduct a task. It is an integral part of the fixed cost of research and development. But we also assume that ‘ a ’ is a function of the level of patent protection, namely $a(\kappa)$, such that $a'(\kappa) > 0$. That is, stronger patent rights work to raise the developmental cost of conducting R&D. There are numerous reasons why. First, they increase the cost of innovation inputs. Under stronger patent protection, input sellers are able to charge higher prices or licensing fees for their technologies. Second, there are upfront transactions costs associated with negotiating rights and cross-licensing agreements, about which stronger patent rights raise the bargaining power of the licensor. Third, the *doctrine of equivalents* puts the onus on the next innovator to clearly differentiate its technology or product quality from the previous generation; the doctrine governs the scope of patent protection. The stronger the patent system, the more widely the doctrine is interpreted such that new quality variations may still fall under the scope of an existing protected innovation. The cost of R&D increases, therefore, as product developers incur more resources and effort to ensure that a new product quality does not infringe upon existing versions. All of these costs are measured in terms of the labor required to conduct the upfront work.¹⁰

To derive an equation for vertical innovation, we require two more pieces: (i) a free entry/exit condition for market entry and (ii) a resource (labor market) constraint. First, the free entry/exit condition. The cost of innovation equals $wa(\kappa)\iota dt$, where w denotes the wage. The expected gain is $V\iota dt$. A finite amount of R&D, ι , occurs if $V = wa(\kappa)$ (or else, ι is either zero or unbounded). In (4), instantaneous profits are given by $\pi = (p - w)x$ (omitting subscripts). The firm charges a markup over marginal cost, $p = \lambda(\kappa)w$, and *limit-prices* the previous innovator out of the market. Hence, we can rewrite instantaneous profits as $\pi = E(1 - \frac{1}{\lambda(\kappa)})$. Using (4), we can write the free

¹⁰See Merges and Nelson (1990) and Merges and Duffy (2017) for a discussion of the doctrine of equivalents. See Glass and Saggi (2002) for a treatment of how IPRs can affect the parameter ‘ a ’ by raising the costs of inventing or imitating around existing quality products. Yang and Maskus (2001) analyze the effects of patent rights on the bargaining position (or rent share) of the licensor. Chu et al. (2012) model an entrant paying a license fee to the incumbent that covers for the degree of infringement.

entry/exit condition as:

$$\frac{E(1 - \frac{1}{\lambda(\kappa)})}{\rho + \iota} = wa(\kappa) \quad (5)$$

Next, the resource constraint is:

$$\bar{L} = a(\kappa)\iota + \frac{E}{\lambda(\kappa)w} \quad (6)$$

where $\bar{L}$ is the total supply of labor. The first term on the RHS is the amount of labor used in R&D and the second term the amount used in production. For the latter, it is assumed that one unit of output requires one unit of labor. The quantity of each product quality is given by $\frac{E}{p}$ and we saw that price is a markup over marginal cost, $\lambda(\kappa)w$, and the total measure of products is 1.

Solving for equations (5) and (6) -- specifically, eliminating wages -- gives us the equation for vertical innovation in steady-state:

$$\iota^* = \frac{\bar{L}}{a(\kappa)}(1 - \frac{1}{\lambda(\kappa)}) - \frac{\rho}{\lambda(\kappa)} \quad (7)$$

We can use equation (7) to determine the impact of patent rights κ on product quality upgrading. First, the quality level of an exportable product is given by $q = \lambda(\kappa)^m$, where $\lambda(\kappa)$ is the size of the jump and m the number of jumps over some time interval $(0, t)$. Second, assume that the jumps followed a *Poisson process* with arrival rate ιt . The mean number of jumps is then $E(m) = \iota t$ over that interval. Thus the quality level of a product is $q = \lambda(\kappa)^{\iota t}$, or:

$$\ln q = \iota^* \ln \lambda(\kappa) \quad (8)$$

where we normalize to the unit time interval. The impact of patent protection on product quality upgrading can be then derived as follows:

$$\frac{\partial \ln q}{\partial \kappa} = \Omega_\lambda \frac{\lambda'(\kappa)}{\lambda(\kappa)} - \Omega_a \frac{a'(\kappa)}{a(\kappa)} \geq 0 \quad (9)$$

The impact is positive if stronger patent protection increases the growth of product quality ($\frac{\lambda'(\kappa)}{\lambda(\kappa)}$) more than it increases the growth of R&D development costs ($\frac{a'(\kappa)}{a(\kappa)}$), and negative otherwise.¹¹ Thus, if rising R&D development costs are onerous and the potential for product quality growth low, then $\frac{\partial \ln q}{\partial \kappa} < 0$. In this case, lowering the level of patent protection, κ , would spur vertical innovation and raise product quality. A less stringent patent system makes it easier for firms to invent around existing product qualities (i.e., the doctrine of equivalents would be interpreted narrowly). The size of the quality jumps would be relatively small and the instantaneous profit flow modest. But coupled with a lower fixed cost of innovation (due to a lowering of $a(\kappa)$), the modest stream of expected profits can support a steady rate of quality upgrading, the kind of incremental improvements that the case of the Chinese ball-pen industry illustrated. A very strong patent system would otherwise raise the entry cost and impede innovation.

In contrast, if the potential for product quality growth is high, a strengthening of patent protection would spur vertical innovation and product quality upgrading, $\frac{\partial \ln q}{\partial \kappa} > 0$. Stronger patent protection would provide incentives for firms to invest in larger product upgrades, and the firms would be willing to incur the higher R&D development costs, given the larger expected firm value upon successful innovation. A weak patent system would otherwise lower the returns to making significant quality improvements. In summary, the analysis suggests that the impact of patent protection on product quality upgrading depends upon the innovation potential of firms. We make the following hypotheses:

H1: Below a threshold level of technological capacity, the relationship between patent protection and quality upgrading is negative.

H2: Above a threshold level of technological capacity, the relationship between patent protection and quality upgrading is positive.

In the empirical analysis, we proxy this level of technological capacity by how close a country's product quality level is to the world frontier.

¹¹The weights are $\Omega_\lambda = \frac{\ln \lambda}{\lambda(\kappa)} (\frac{\bar{L}}{a(\kappa)} + \rho) + \iota^*$ and $\Omega_a = \frac{\bar{L} \ln \lambda}{a(\kappa)} (1 - \frac{1}{\lambda(\kappa)})$.

4 Data

The index of export product quality is from the International Monetary Fund (IMF).¹² Its product quality data are based on revision 1 of the *Standard International Trade Classification* (SITC). The IMF quality index is quite extensive, covering 178 countries and hundreds of products during the period 1962 - 2010. The IMF quality index is reported at the SITC 1-, 2-, 3-, 4-digit product levels, as well as at the aggregate country-level. This paper uses the most detailed SITC 4-digit level data on manufacturing product quality.¹³ Quality is measured as the unit value, adjusted for differences in production costs and for the selection bias arising from relative distance. Specifically, unit value is a function of quality, its production cost (or the exporter's GDP per capita), and distance. Higher quality, higher production costs, and longer distance to delivery result in higher prices, and vice versa. Hence, quality can be pulled from unit prices, exporter's GDP per capita, and distance. A quality-augmented gravity equation is used to estimate the coefficients of the quality determinants. Henn et al. (2017) make the case that the IMF quality index is more efficient than the unit value approach in identifying unobserved product quality, especially when a country's quality level reaches about 80 - 85% of that of the world frontier. Increases in quality beyond that level tend not to be associated with price increases, possibly because higher efficiency in production reduces costs.

Manufacturing products are either quality heterogeneous or quality homogeneous. Given our interest in vertical innovation, we focus on quality heterogeneous products and exclude quality homogeneous products. The trim is based on Rauch (1999), which classifies products as quality differentiated or non-differentiated.¹⁴ The Rauch (1999) classification of quality differentiated products follows SITC revision 2.¹⁵ We therefore concorded the IMF product quality index, as

¹²From <http://www.imf.org/external/np/res/dfidimf/diversification.htm>.

¹³Manufacturing products consist of SITC product groups 5 to 8.

¹⁴Specifically, Rauch (1999) classifies products into homogeneous, reference priced, and differentiated categories. Bastos and Silva (2010) show that Rauch's differentiated products are well suited for capturing quality differentiation. Bas and Strauss-Kahn (2015) also utilize a similar procedure.

¹⁵See http://econweb.ucsd.edu/~jrauch/rauch_classification.html.

reported in SITC revision 1, to the SITC revision 2 classification, and then kept those quality differentiated manufacturing products as classified by Rauch (1999) in the dataset.¹⁶

Since it takes time for export product quality to grow through either imitation or innovation, and time for technologically capable enterprises to react to patent reforms, we retain only those countries and products with continuous export records between 1990 and 2010.

Our key explanatory factor is the strength of patent protection. We use an index of strength that is based on the legal provision of rights and exceptions to patent holders. The index consists of five components, each of which is scored out of one: first, the duration of patent protection relative to the international standard; second, subject matter that is patentable (or not unpatentable); third, participation in international IPR agreements; fourth, enforcement mechanisms available; and fifth, how limited (or less restricted) the patenting exceptions are (such as any requirement to practice the invention or license the patents to third-parties). The overall index of patent protection varies from zero to five, with higher numbers reflecting strong levels of patent rights.¹⁷

However, in measuring the level of patent protection, it is important to account for the enforcement of legal provisions in practice. We therefore follow Hu and Png (2012) (and subsequently by Maskus and Yang, 2018) and utilize an index of legal enforcement effectiveness developed by the *Fraser Institute* to adjust the index of patent protection. Specifically, effective patent rights, PRI , in country i at time t is given by:

$$PRI_{it} = \theta_{it} \times \text{Patent Rights Index}_{it} \quad (10)$$

where $0 \leq \theta_{it} \leq 1$ is the Fraser index of legal enforcement effectiveness. θ downward adjusts the measured index of patent protection for imperfect enforcement. The Fraser index of legal enforcement effectiveness ranges from zero to ten. We scaled it so that it varies from zero to one; i.e., deflated the Fraser index by 10. If enforcement effectiveness is at a maximum, then $\theta = 1$ and

¹⁶The concordance table is from the UN Statistics Division.

¹⁷The source of data is Park (2008).

PRI = Patent Rights Index. If enforcement is non-existent, then $\theta = 0$ and $PRI = 0$. The Fraser index is based on three aspects of legal enforcement: legal security from confiscation of property rights, viability of contracts, and the rule of law.¹⁸ The Fraser index is a subjective assessment, as it is compiled from surveys of international business executives. Thus PRI should capture the scope of effective IPR protection (Maskus and Yang, 2018).

Another key variable of interest is a measure of the *stage* of technological development. The stage will vary by product and country; that is, a country may be relatively higher up the technology ladder in certain goods than in other products. We thus measure the closeness to the world frontier, CTF , for each country and product combination. CTF is calculated as follows:

$$CTF_{ij} = \left(\frac{QLTY_{ij}^{initial} - QLTY_{lowest,j}^{initial}}{QLTY_{highest,j}^{initial} - QLTY_{lowest,j}^{initial}} \right) \times 100 \quad (11)$$

where $QLTY_{ij}^{initial}$, and so forth, are the initial values; that is, we use pre-sample data (averages of 1985 - 1990) to construct the initial stages of technological development. This allows us to test the predicted effects of patent rights reforms on the subsequent shifts in export product quality, according to what the levels of technological development were prior to the reforms. Note that when calculating the world's highest quality ($QLTY_{highest,j}^{initial}$) and world's lowest ($QLTY_{lowest,j}^{initial}$) quality for product j we examined all countries in the database, including advanced countries. The formula (11) borrows from the 'Distance to Frontier' methodology underlying the World Bank *Doing Business* dataset, except that we take the inverse so that higher values of CTF reflect higher positions in the technology ladder.¹⁹ The values of CTF_{ij} range from zero to 100, where 'hundred' signifies that a country-product is right at the world frontier.

In the absence of data on product-level labor productivity, previous studies have employed total factor productivity (TFP) as a proxy for the level of technology (e.g., Acemoglu et al., 2006). Here

¹⁸Available at http://www.freetheworld.com/datasets_efw.html.

¹⁹Details of the methodology can be found at <http://www.doingbusiness.org/data/~media/WBG/DoingBusiness/Documents/Annual-Reports/English/DB18-Chapters/DB18-DTF-and-DBRankings.pdf>.

we believe that product quality itself is a natural proxy for product-level technology. In previous research, the measure of distance to frontier has been constructed in multiple ways. Chu et al. (2014) employ country-level data from the *Penn World Tables* on labor productivity relative to the U.S. as an inverse measure of the distance to frontier; however, this does not allow for differences across products. Acemoglu et al. (2006) defines industry-level “proximity to technology frontier” as the level of TFP per industry, country, and time, divided by the highest TFP per industry and time in the sample. Our approach allows for more disaggregation at the product level. Amiti and Khandelwal (2013) is the first to construct a detailed product-level measure of proximity to technology frontier by using their self-developed product-level quality index divided by the highest quality level within each HS ten-digit product. Rather than take the ratio of a product’s quality to the world maximum, we use a instead different formula for distance to frontier, as shown in (11).

An advantage of the formula in (11) is that it can quantitatively depict the position in the technology ladder – namely, the position between the world’s lowest quality index and world’s highest index; for example, a value of $CTF_{ij} = 50$ indicates that the quality of product j of country i is halfway between the world’s leading brand and the world’s lagging.²⁰ Another difference with Amiti and Khandelwal (2013) is that we base our quality index on world exports rather than on exports to the U.S. only. This allows us to include more products of developing countries than if we just concentrate on a market like the U.S.²¹

Overall, our sample consists of 46 developing economies (see Appendix I). Our classification of developing countries is based on the World Bank’s classification of non high-income economies. We also obtain 235 SITC 4-digit quality differentiated products. Appendix II shows a list of industries covered in our study. Although we utilize data at the detailed four-digit SITC level, Appendix II clusters them into 23 two-digit SITC sectors so as to conserve space. Overall, we

²⁰Later in a robustness test we show how our results are affected if we use the ratio of product quality to the world maximum as a measure of the closeness-to-frontier.

²¹For example, China has been the world’s largest automobile market since 2011 in terms of both supply and demand (see <https://www.cnbc.com/2011/09/12/Worlds-10-Largest-Auto-Markets.html>). But until recently, China did not export passenger cars to the U.S., only auto parts. Hallak (2006) and Feenstra and Romalis (2014) also use exports to the world rather than exports to the U.S. only.

have 18,245 country-product observations in the dataset from 1990 to 2010 at 5-year intervals. In our sample, Brazil has the largest number of continuously exported goods, exporting 218 SITC 4-digit quality differentiated products, which accounts for 5.97% of the total country-product observations, followed by China exporting 216 SITC 4-digit products. Machinery, other than electrical, dominates the share of exported products, as exported by 38 of the countries in the sample.

For control variables, we include gross domestic product per capita (GDP_PC) in constant 2005 U.S. dollars, FDI net inflows as a % of GDP (FDI), the percentage of complete secondary school attained in the total population as a measure of human capital (HC), and the most favored nation applied tariffs ($TARIFF$). Appendix III summarizes our data sources and Table 1 provides some descriptive statistics.²²

5 Empirical Methodology

Based on the conceptual framework, we estimate the following model:²³

$$QLTY_{ijt} = \alpha_i + \alpha_j + \alpha_t + \beta_1 PRI_{it} + \beta_2 PRI_{it} \times CTF_{ij} + \beta \mathbf{X}_{ijt} + \epsilon_{ijt} \quad (12)$$

where the subscripts i , j , and t index country, product, and year respectively. $QLTY$ denotes export product quality, PRI the measure of effective patent rights, and CTF a measure of the closeness-to-the-world technological frontier. All three variables, $QLTY$, PRI , and CTF are in natural logarithms. $\mathbf{X}$ represents a vector of control variables that includes GDP per capita (in logs), import tariff rate, and the rate of human capital and FDI inflows. The α_i , α_j , and α_t , represent country, product, and year fixed effects, which are included to control for any omitted variable biases. As Henn et al. (2017) point out, country fixed effects can control for the possibility

²²The data that support the findings of this study are from publicly available sources. The compiled dataset can be obtained from the authors upon request.

²³The equation is estimated as a linear regression with high-dimensional fixed effects.

that quality growth is faster in some countries, owing to unobserved institutional circumstances whereby certain business organizations are better able to exploit knowledge spillovers. Product fixed effects help control for quality improvements being easier to attain in some products. Year fixed effects control for changes over time in the global average speed of quality growth, due for instance to advances in information and communications technology or reductions in transportation costs. Note that CTF_{ij} varies only along the country-product dimension.

In (12), we follow Maskus and Yang (2018) and lag the effective patent protection variable, PRI , by one period (or five years) to take into account that patent policies take some time to influence investments in product quality. For consistency we lag other control variables in the same way. Such a strategy combined with the inclusion of fixed effects in the regression, as Maskus and Yang (2018) argue, can largely mitigate the potential endogeneity concerns.²⁴ In section 6, we confirm with an endogeneity test that our results do not suffer from a simultaneity bias.

$PRI \times CTF$ is the interaction between effective patent protection and the closeness-to-the-world technological frontier. It is the key term for measuring the stage-dependent effect of patent rights on export quality. CTF is set to be time-invariant. The reason is two-fold. First, we want to condition the effects of patent protection on export product quality on the technological stage of development. The period under investigation (1990 - 2010) was a time of major changes in IP protection for developing economies. Our framework predicts that the benefits to product quality upgrading depends on whether countries and products met a critical threshold level of closeness to frontier. Second, we want to avoid endogeneity, whereby technological distance responds to policies and investments in product quality. By fixing the value of CTF for each product-country pair over the entire sample period, we can more sharply measure the conditioning role of distance-to-frontier on our relationships of interest.

²⁴Delgado et al. (2013) argue that the TRIPS agreement constituted a sort of ‘natural experiment’ in that most countries (especially developing) were reluctant to join it. Their membership in the agreement was not likely an endogenous response to shifts in policy or innovative capacity. Delgado et al. (2013) find no pre-TRIPS patterns that suggest a major endogeneity issue. The authors argue that their “differences-in-differences approach allows [them] to address unobserved country-level changes that might also drive the adoption of IPRs.”

Differentiating the vertical innovation measure with respect to effective patent protection in equation (12) gives us:

$$\frac{\partial QLTY_{ijt}}{\partial PRI_{it}} = \beta_1 + \beta_2 CTF_{ij} \quad (13)$$

From our conceptual framework, our hypotheses are that $\beta_1 < 0$ and $\beta_2 > 0$. That is, if the quality level of a country-product combination is far away from the world technology frontier (i.e., a small CTF_{ij}), the effect of PRI on export quality would be negative. If the quality level of a country-product combination is close to the world technology frontier (i.e., a large CTF_{ij}), the effect of PRI on export quality would be positive. We can derive the critical value of the closeness-to-the-world technological frontier from:

$$CTF_{ij}^{\text{threshold}} = -\frac{\beta_1}{\beta_2} \quad (14)$$

which we will compute (by taking the exponential of the RHS of (14)).

6 Results

6.1 Preliminary Analyses

Through some sample statistics, we can observe signs of a stage-dependent effect. That is, in the raw data, we can see some correlations between export product quality and effective patent protection that vary according to whether a country and product mix is above or below some threshold closeness-to-the-world frontier. In Table 2, we split the sample into four groups. Each group can be associated with a country that has either above-median effective patent protection or below-median effective patent protection. Each group can also consist of a product-country closeness-to-the-world frontier value that is in either the bottom quartile or top quartile. The reason we cut the closeness-to-frontier measure into quartiles rather than above/below median CTF is that we do not yet know the threshold value of CTF (as we will later estimate with our regression models). For each of the four possible cases, Table 2 reports the mean value of the

index of export product quality for that combination (along with the number of observations). The reason we observe signs of a stage-dependent effect is that in the bottom quartile of the closeness-to-the-world frontier – that is, when technological levels are ‘low’ – the mean export product quality is higher if patent protection is effectively lower (i.e., if *PRI* is below the sample median). In contrast, in the top quartile levels of *CTF*, the mean export product quality is higher if patent protection is effectively higher (i.e., if *PRI* is above the sample median). The average quality differences seem quantitatively small, but they are statistically significantly different (at the one percent level). These findings are consistent with our hypotheses that stronger effective patent protection would be associated with higher export product quality if the technological level (as captured by a country-product’s closeness to the world frontier) is sufficiently high, but if the technological level is low, less stringent patent rights would be associated with higher export product quality.

Table 3 puts a time dimension to what we have analyzed in Table 2, showing that export product quality and effective patent protection vary positively between 1990 and 2010 for upper middle-income countries, while they vary inversely over time for lower middle-income and low-income countries. As Table 3 indicates, upper middle-income countries are on average closer to the world frontier than are lower middle-income countries, who in turn are closer to it than the low-income countries.²⁵ As the last column of Table 3 shows, between 1990 and 2010, effective patent protection levels more than doubled for all income groups, but the average export product quality index fell, except in the upper middle-income group. These trends again are consistent with our hypotheses that stronger effective patent protection would deter export quality upgrading if the products of countries are low in the world quality ladder. Of course, a limitation with looking at the raw data like this is that Table 3 does not control for other changing factors.

²⁵Note that we divided each income group’s *CTF* by the low-income group’s *CTF* so as to normalize it to the latter. The classification of countries by income group is based on the World Bank’s classification.

6.2 Regression Results

We next examine our hypotheses in a more controlled setting. Table 4 contains our main results. Consider first the pooled sample (see column (1)). The coefficient of the interaction term ($PRI \times CTF$) is positive and significant at the 1% level, and the coefficient of effective patent protection PRI is negative and also significant at the 1% level, meaning that unless technological potential, as captured by CTF , is sufficiently high, stronger patent rights are not conducive to upgrading export product quality.

Based on the two estimated coefficients in column (1) ($\hat{\beta}_1 = -0.097$, $\hat{\beta}_2 = 0.027$), the threshold value for the stage-dependent effect is $36.3 (= e^{(0.097/0.027)})$, using the formula derived in (14). This suggests that if the technological level of a country's product is below 36.3% of the world frontier, effective patent protection is negatively associated with export product quality. But if it is more than 36.3% close to the world level, effective patent protection would be positively associated with export product quality. Thus, the lower the threshold value or cutoff is, the greater the range of product qualities (from the top of the quality ladder down to the threshold) that could improve from stronger patent protection.

Next, we split the sample into different groups and find some heterogeneity in results. In column (2) of Table 4, we show the estimates of the model for the low income countries. This group tends to produce goods that are at the lowest end of the quality ladder. As shown in column (2), the estimated coefficient of effective patent protection is $\hat{\beta}_1 = -0.079$ and is significant at the 10% level. However, the estimated coefficient of the interaction term is positive (i.e., $\hat{\beta}_2 = 0.020$) but is not statistically significant. This implies that strengthening patent protection in low-income countries primarily hinders their export quality upgrading. In this case, there is no apparent stage-dependent effect, or no turning point for the effects of patent rights.

The lower middle and upper middle income countries do specialize in goods that are at a higher end of the quality ladder than what the low income economies produce. Thus, in columns (3)

and (4) where we examine these groups respectively, we do find the estimated coefficient of the interaction term to be both positive and significant at the 1% level. Using (14), we find the estimated threshold cutoffs to be 37.5 for the lower middle income group and 42.1 for the upper middle income group. Above these thresholds, a strengthening of effective patent protection would spur export quality upgrading. But note that the range of product qualities that could be spurred by stronger patents is narrower for the upper middle income countries than for the lower middle income countries. Among the upper middle income group, those country-product qualities within 57.9% of the world's apex quality would profit from stronger effective patents, whereas among the lower middle income group, those within 62.5% can. One possible reason has been explored in Shin et al. (2016), which argued that if upper middle income countries produce goods that are at the relatively higher end of the world's quality ladder, producers there encounter vigorous competition with exporters from advanced economies. Consequently, in order to benefit from stronger effective patent rights, the required technological potential of exporters in upper middle income countries should be greater than that of exporters in lower middle income countries.

Next we split the sample by product groups. In columns (5) and (6), we estimate the model on a sample of high intellectual property (IP) products and non-high IP products respectively. The classification of products into high IP versus the rest is based on Delgado et al. (2013).²⁶ In both the high IP group and non-high IP group, we detect a stage-dependent effect: the estimated coefficient of *PRI*, the effective patent protection variable, is significantly negative, while the estimated coefficient of the interaction between *PRI* and *CTF*, the closeness-to-frontier, is significantly positive. The estimated threshold level is 49.1 for high IP products and 35.4 for the rest. The reason the threshold value is higher for high IP products is that the quality differentiation needs are greater so as to avoid stepping upon (or infringing upon) existing product qualities. Thus stronger effective patent protection would spur the upgrading of high IP exports if the technological potential is sufficiently high or the distance-to-the-world frontier is sufficiently close.

²⁶We follow Table 4Aa of their online appendix. High IP products include office machines, chemicals, power generating machinery, professional apparatus, telecommunications, and so forth. Their definitions are based on SITC rev. 3, which we concurred to our version.

We repeat the above exercise using another way to measure the IP contents of products. Following Hu and Png (2012), we classify products according to whether they are highly patent intensive. Patent intensity is measured as the number of U.S. patents granted to an industry relative to total industry sales in the U.S. We matched the industries to our product classes. Patent intensity is coded as low if it is below the median value and as high if it is above the median. The results appear in the last two columns of Table 4 and are consistent with the findings in columns (5) and (6). In each sub-sample of products, we find a stage-dependent effect and find that the threshold value is higher for high patent intensity products (as reported in the last row of Table 4).

As for the control variables, GDP per capita has a positive association with the upgrading of export product quality, which is consistent with the findings in the previous literature.²⁷ The human capital and FDI measures have a statistically insignificant association with export product quality, except for low-income countries. This underscores the importance of importing foreign knowledge and overcoming educational barriers in regions lacking technological capacities (Chen and Puttinanum, 2005, Chu et al., 2014, and De Fuentes et al., 2018). Henn et al. (2017) find the effects of human capital and FDI to be statistically significant but economically negligible. Tariffs have a positive association with export quality in some cases and an insignificant one in others. Prior research has identified ambiguous effects. For instance, firm level evidence in Bas and Strauss-Kahn (2015), Fan et al. (2015), and Liu et al. (2019) show that tariff reductions can promote export quality upgrading by allowing higher quality inputs to be imported or by increasing competition. Multi-country studies, such as Amiti and Khandelwal (2013), show that if product quality is far from the frontier, tariff reductions could discourage lagging firms from undertaking product innovation, given their inability to cope with increased competition.

²⁷For example, Schott (2004), Khandelwal (2010), and Henn et al. (2017).

6.3 Robustness Check: Alternative Measures of CTF and PRI

We undertake a series of robustness checks by using a variation of our measures of the closeness-to-the-world frontier and of effective patent protection. Instead of (11), we can simply measure CTF as the ratio of a product’s quality to the highest quality within each SITC 4-digit product class; this is the approach in Amiti and Khandelwal (2013). And instead of (10), we can dispense with the adjustment for enforcement effectiveness and examine the index of patent rights as is. The results are reported in Table 5 for the pooled sample only. In this table, we denote CTF^{Alt} as the natural log of the *alternative* measure of closeness-to-frontier (namely, $CTF_{ij}^{Alt} = \frac{QLTY_{ij}^{initial}}{QLTY_{highest,j}^{initial}} \times 100$) and PRI^{Alt} as the natural log of the index of patent rights (without adjustment for enforcement effectiveness).

There are three columns in Table 5. In column (1), we apply CTF^{Alt} ; in column (2), PRI^{Alt} ; and in column (3), both CTF^{Alt} and PRI^{Alt} . As column (1) shows, the sign patterns and the significance of the coefficients of effective patent protection and its interaction with the alternative measure of technological closeness fit our hypotheses. Using this alternative CTF , the threshold value at which the effect of patent rights on export product quality switches from negative to positive is 76.8 ($=e^{(0.864/0.199)}$). In other words, a country and product combination needs to have a quality level that is at least 76.8% of the world’s highest quality for that product line in order for a strengthening of patent enforcement to encourage the upgrading of export quality. This is much higher than the 36.3% estimate that we obtained earlier. Thus it appears that while taking the ratio of a product’s quality to the world’s highest quality gives us similar qualitative conclusions about the efficacy of effective patent protection, the alternative method underestimates the potential impact of IPR reforms. We say ‘underestimate’ in the sense the alternative measure of CTF reduces the range of quality levels at which IPR reforms can spur export quality upgrading; only the very high end of the distribution – namely the top 23.2% of the quality distribution – can prosper from stricter patent enforcement. The export upgrading of the bottom nearly three-quarters of the quality distribution is predicted to be hampered by stricter patent enforcement.

We favor our approach because it takes into account not just the quality leaders but the lagging qualities with which export products compete.

In column (2), where we do not adjust patent protection for enforcement effectiveness, we again find a stage-dependent effect whereby stronger patent protection is associated with lower export product quality for countries and products whose quality levels are far from the world frontier, and the reverse for quality levels closer to the world frontier. The threshold value is 49.4. This is closer to what we have obtained before (for the pooled sample), but the fact that it is a slightly larger value suggests that taking into account enforcement effectiveness provides more scope around the margin for patent rights to spur export quality upgrading, namely for product qualities just under the middle of the world distribution.

In column (3), where we use both the alternative measures, we again confirm the hypothesized stage-dependent effect. The threshold cutoff is 79.0. This threshold is also very high relative to what we found earlier and shows a more limited role for patent protection to spur the upgrading of export product quality. For this reason, we prefer our approach to computing CTF in (11), which finds the relative position between the highest and lowest existing values of CTF . The alternative, CTF^{Alt} , assumes the lowest value is zero.

6.4 Robustness Check: Sample Sensitivities

One potential concern is that the highest or lowest quality products may influence the result. Therefore, we exclude product qualities below the 5th percentile and above the 95th percentile of the quality index. We re-run the regression for the pooled sample. The results are reported in Column 1, Table 6. The threshold value is 35.0, which is very close to the finding in Table 4, column 1. Hence, the top and bottom five percents do not constitute severe outliers.

Another concern is the bias that may arise from the inclusion of the export quality index for China. Schott (2008) analyzes the dramatic growth of Chinese goods in world product markets,

but finds big discounts in Chinese export prices, suggesting that China exports relatively less sophisticated varieties. In a similar vein, Amiti and Khandelwal (2013) argue that China’s export quality may be overstated because the U.S. imports data report gross values rather than value added. This concern may also apply to the methodology adopted in Henn et al. (2017). In our dataset, China ranks second in terms of the number of observations, accounting for almost 6.0% of the total observations.

Column 2, Table 6 shows the result of excluding China from the sample. Again our stage-dependence hypothesis still holds, and the calculated threshold value is 38.0 which is also close to the earlier finding.

6.5 Test of Endogeneity

This section addresses the potential endogeneity between export product quality and patent protection. The decision to strengthen patent rights could have been driven by the growth in export capacity and quality, as this may have raised the demand for intellectual property protection by innovating industries. The techniques employed in the paper thus far – lagging the index of PRI and including country and product fixed effects – mitigate these concerns but do not fully correct for endogeneity. To estimate the causal effect of patent protection, we adapt the IV approach of Ivus (2010) in which colonial origin is used to isolate exogenous variation in patent rights. Ivus (2010) used this approach to test for the endogeneity between PRI and export quantity (in terms of export value). We apply the same test for export *quality*.

The logic behind colonial origins as an instrument is as follows: among developing countries, we can observe diverse patterns in the strengthening of PRI. Prior to TRIPS, the former colonies of Britain and France had higher levels of patent protection than did the non-colonies, and experienced increases in the level of PRI. But post-TRIPS, it was the non-colonies that had the most significant rise in PRI. Thus in the pre-TRIPS period, the patent rights of colonies increased most likely due

to their relations with France and the U.K. That is, the colonizers influenced the institutions of their colonies. This is further supported by Zorina Khan (2004) who argued that France and the U.K. had well-developed systems of intellectual property protection which influenced the systems of the colonies during the 19th century. This is also corroborated in the design of plant variety protection in which the colonies of France and the U.K. fashioned their systems after France and the U.K. respectively (Campi and Nuvolari, 2015).

Thus, the level and growth of PRI for the colonies had their origins pre-TRIPS. In contrast, the strengthening of PRI for non-colonies after 1990 was mostly likely a consequence of TRIPS. Indeed, a dramatic increase in PRI can be observed among the non-colonies post-TRIPS relative to pre-TRIPS. TRIPS had mandated the non-colonies (who were signatories to TRIPS) to raise their standards for patent protection. For this reason as argued in Delgado et al. (2013) TRIPS can be thought of as exogenous to the non-colonies (the treatment group), whereas the former colonies of the U.K. and France can be classified as the non-treated group. We expect PRI to affect the treated group (non-colonies).

To implement the IV approach, we difference the data over the 20-year period (1990 - 2010) and relate changes in export quality to changes in patent rights. The resulting data are a cross-section of country and product combinations. Among the 46 developing countries in our sample, 20 are former colonies and 26 are non-colonies. We use two variables – a non-colony dummy and the interaction of the non-colony dummy with the closeness-to-frontier measure (CTF) – as excluded instruments for the two endogenous variables, ΔPRI_{it} and $\Delta PRI_{it} \times CTF_{ij}$.

Table 7 shows the IV estimates. Columns (1)-(2) report the Stage 1 results, where each of the two endogenous variables are the outcome variables, and column (3) reports the Stage 2 results, where $\Delta QLTY_{ijt}$ is the outcome variable. The first stage results indicate that the non-colony variable is a significant determinant of changes in patent rights, and the interaction of the non-colony variable with closeness-to-frontier is also a significant determinant of the interaction between changes in patent rights and closeness-to-frontier. The test of underidentification rejects the null hypothesis

of underidentification at the 0.001% level and indicates that these instruments are relevant. The Weak Identification test suggests that the instruments are not weak. The endogeneity test indicates that we cannot reject the null hypothesis of the exogeneity of patent rights changes, ΔPRI , and the exogeneity of their interaction with closeness-to-frontier, CTF . Thus, the results we have reported thus far do not appear to exhibit any endogeneity bias.

6.6 Discussion: Policy Implications

Thus far, we have been showing estimates of the closeness-to-frontier threshold CTF , above which patent protection is positively associated with export quality grading and below which it is negatively associated. But what is the density of the distribution of CTF below and above that threshold? This is important for knowing whether on balance stronger effective patent protection spurs or hinders export quality upgrading for most of the products and economies in the sample. For example, if most of the country-products are below the threshold, patent reforms would overall impede vertical innovation in the developing world.

Recall from (11) that CTF ranges from zero to 100. In Table 8, we distribute all the CTF values in our sample into ten ranges (or deciles); for example, 0 to 10, 10 to 20, ..., 90 to 100. We especially take note of our main threshold value of 36.3% which was derived from the pooled sample and is marked in bold in Table 8. For each range, we show the total count of country-product pairs or combinations that fall in that range. We further break that count down by income group: upper middle income countries, lower middle income countries, and low income countries. Not surprisingly, products in the low income group are heavily concentrated in the lowest ranges of closeness-to-frontier (see the first three deciles). Indeed there are no country-product counts in the 80 to 100 range for low income countries. In contrast, most of the country-product combinations in the higher ranges of CTF – from 50 to 100 – belong to the upper middle income developing countries.

For the pooled sample, this table seems to show that stronger effective patent protection would spur export upgrading for the majority of country-product combinations. According to column (2) of Table 8, 75% of country-product combinations are above the critical value of the closeness-to-the-world frontier. This is the range of cases where stronger patent rights would facilitate vertical innovation. But this is a somewhat inflated share since we need to take into account group-specific thresholds (as shown in Table 4). When we split the sample by income group, it is clear that stronger effective patent protection would largely profit exporters in the upper middle income countries; for this group of countries, 80.6% of country-product pairs are above their within-group threshold quality. For the lower middle income group and low income group, the share of country-products above their respective quality thresholds are 50.7% and 0% respectively. Consequently, if we look at the counts of country-product combinations broken down by income group, we find 2,528 combinations are above the threshold and 1,121 below.²⁸ Thus, patent reforms would spur export quality growth for 69% (not 75%) of country-product combinations. So, while the majority of cases reveals a positive impact of patent rights on export quality upgrading, the enhancements occur mostly in the upper-income developing economies.

What makes patent policy recommendations complex is that we cannot simply make a blanket recommendation that patent rights and enforcement be strengthened in upper middle income countries and relaxed in lower income nations. The reason is that the unit of analysis is the country-product pair. The same product can be on different sides of the threshold in different countries. For example, pharmaceutical products in Brazil were above the threshold (closeness-to-frontier = 65.3) but below the threshold in India (closeness-to-frontier = 29.4), which is interesting given India's strong generic drug industry. Chemical products and preparations were above the threshold in South Africa (66.5) but below in China (27.5). And within the same country, different products can be on different sides of the threshold. To hark back to our case study, we find that Chinese ball-point pens were above the threshold (44.7), making it ripe for quality upgrading under

²⁸In Table 8, the row for 'above threshold' shows 2,002 country-products from the upper middle income group and 526 from the lower middle income group, or 2,528 in total. The total below the threshold is the balance of the overall count of 3,684 minus 2,528.

patent reforms, while Chinese chemical products were, as we saw, below the threshold. In India, pens were also above the threshold (41.9). This raises the question of whether there should be product or industrial differentiation in the design of patent policy. Currently, across the world, patent laws are national; that is, with few exceptions, they apply uniformly to different industries and/or products. Future research and policy debates could consider IP regimes that differentiate the strengths of protection by industry, product, or technology.²⁹

7 Conclusion

This paper contributes new insights into the export quality literature. While there has been vibrant research on the determinants of product quality upgrading, no study has yet explicitly examined the impact of IPR policy on quality upgrading, as far as we are aware. As Delgado et al (2013) argue, many developing economies were obligated to undertake patent reforms. Empirical research has actively sought evidence on the impacts of these reforms, be they on innovation, economic growth and development, or participation in international trade. Our study shows that the effects of effective patent protection on export quality are not unambiguous but conditional on the stage of the technological development of countries and products. As we had hypothesized, a strong effective patent system rewards innovators capable of producing sufficiently large quality jumps and hinders innovation targeted at incremental jumps.

Research by Chen and Puttitanun (2005), Chu et al. (2014), among others, has analyzed a stage-dependent effect of IPRs on economic development and growth, but to date no formal evidence has been produced to show that the effects of IPR protection on export product quality vary by stage, particularly for the developing countries that have had to undertake sweeping IPR reforms. By employing the most extensive IMF product quality index thus far and data on effective patent protection during two decades, this paper fills a gap in the literature and produces direct

²⁹Cho et al. (2015) discuss the possibility of differentiating IP standards by industry and Acemoglu and Akcigit (2012) argue for different protection levels for industry leaders and followers.

product-level evidence of a technological stage-dependent effect.

Our results should have relevance for policy. First, for developing countries, enforcing identical standards of patent protection across countries and/or industries is not desirable for promoting the growth of export product quality. The patent protection levels implemented by policymakers should be differentiated across products by their closeness to the world's technological frontier; for example, different patentable subject matter should be treated differently, such as software, pharmaceuticals, and machinery, where their levels of product sophistication can vary considerably across regions. Furthermore, countries that are farther from the world frontier could adopt more flexible limitations and exceptions to patent rights, such as requiring firms to practice (or work) their patented inventions rather than hold them merely to exclude rivals from entering their markets. As another example, firms in emerging economies that are in the learning and adaptation stages could benefit from a more flexible grace period, rather than a one-size-fits-all term. This is the period during which public disclosure of an invention does not void a patent grant because the invention is no longer novel (since the public knows about it). Typically countries offer a grace period of either 6 months or a year. Innovators of products with quality levels that are far from the world frontier may require a longer grace period since public demonstrations, exhibitions, and testing over time could provide useful feedback for product development.

References

- [1] Acemoglu, Daron and Akcigit, Ufuk (2012), “Intellectual property rights policy, competition, and innovation.” *Journal of the European Economic Association*, 10(1), 1-42.
- [2] Acemoglu, Daron, Zilibotti, Fabrizio, and Aghion, Philippe (2006), “Distance to frontier, selection, and economic growth.” *Journal of the European Economic Association*, 4(1), 37-74.
- [3] Akcigit, Ufuk, Ates, Sina T., and Impullitti, Giammario (2018), “Innovation and trade policy in a globalized world.” Center for Economic Policy Research, Discussion Paper No. 12888.
- [4] Amiti, Mary and Khandelwal, Amit. K. (2013), “Import competition and quality upgrading.” *Review of Economics and Statistics*, 95(2), 476-90.
- [5] Anwar, Sajid and Sun, Sizhong (2018), “Foreign direct investment and export quality upgrading in China’s manufacturing sector.” *International Review of Economics and Finance*, 54, 289-298.
- [6] Baldwin, Richard and Harrigan, James (2011), “Zeros, quality, and space: trade theory and trade evidence.” *American Economic Journal: Microeconomics*, 3(2), 60-88.
- [7] Bas, Maria and Strauss-Kahn, Vanessa (2015), “Input-trade liberalization, export prices, and quality upgrading.” *Journal of International Economics*, 95, 250-262.
- [8] Bastos, Paulo and Silva, Joana (2010), “Identifying vertically differentiated products.” *Economics Letters*, 106(1), 32-34.
- [9] Campi, Mercedes and Nuvolari, Alessandro (2015), “Intellectual property protection in plant varieties. A worldwide index (1961-2011).” *Research Policy*, 44(4), 951-964.
- [10] Chen, Yong Min and Puttitanun, Thitima (2005), “Intellectual property rights and innovation in developing countries.” *Journal of Development Economics*, 78(2), 474-493.
- [11] Cho, Kyungchul, Kim, Changseok, and Shin, Juneseuk (2015), “Differential effects of intellectual property rights on innovation and economic performance: A cross-industry investigation.” *Science and Public Policy*, 42, 827-840.
- [12] Chu, Angus, Cozzi, Guido, and Galli, Silvia (2012), “Does intellectual monopoly stimulate or stifle innovation?” *European Economic Review*, 56, 727-746.
- [13] Chu, Angus, Cozzi, Guido, and Galli, Silvia (2014), “Stage-dependent intellectual property rights.” *Journal of Development Economics*, 106, 239-249.
- [14] De Fuentes, Claudia, Santiago, Fernando, and Temel, Serdal (2018), “Perception of innovation barriers by successful and unsuccessful innovators in emerging economies.” *Journal of Technology Transfer*, forthcoming.
- [15] Delgado, Mercedes, Kyle, Margaret, and McGahan, Anita M. (2013), “Intellectual property protection and the geography of trade.” *Journal of Industrial Economics*, 61(3), 733-762.

- [16] Fan, Hai, Chao, Li, Yao, Amber, and Yeaple, Stephen R. (2015), "Trade liberalization, quality, and export prices." *Review of Economics and Statistics*, 97(5), 1033-1051.
- [17] Feenstra, Robert, C. and Romalis, John. (2014), "International prices and endogenous quality." *Quarterly Journal of Economics*, 129(2), 477-527.
- [18] Glass, Amy and Saggi, Kamal (2002), "Intellectual property rights and foreign direct investment." *Journal of International Economics*, 56, 387-410.
- [19] Grossman, Gene and Helpman, Elhanan (1991), "Quality ladders in the theory of growth." *Review of Economic Studies*, 58, 43-61.
- [20] Hallak, Juan Carlos (2006), "Product quality and the direction of trade." *Journal of International Economics*, 68(1): 238-265.
- [21] Harding, Torfinn and Javorcik, Beata S. (2012), "FDI and export upgrading." *Review of Economics and Statistics*, 94(4), 964-980.
- [22] Henn, Christian, Papageorgiou, Chris, Romero, Jose Manuel, and Spatafora, Nikola (2017), "Export Quality in Advanced and Developing Economies: Evidence from a New Data Set." Policy Research Working Paper No. 8196, World Bank, Washington, DC.
- [23] Hu, Albert and Png, Ivan (2012). "Patent rights and economic growth: evidence from cross-country panels of manufacturing industries." *Oxford Economic Papers*, 65(3), 675-698.
- [24] Hwang, Hong, Wu, Jollene Z., and Yu, Eden S. H. (2016), "Innovation, imitation and intellectual property rights in developing countries." *Review of Development Economics*, 20(1), 138-151.
- [25] Ivus, Olena (2010), "Do Stronger Patent Rights Raise High-Tech Exports to the Developing World?" *Journal of International Economics*, 81(1), 38-47.
- [26] Ivus, Olena and Park, Walter G. (2019), "Patent reforms and exporter behavior: firm-level evidence from developing countries." *Journal of the Japanese and International Economies*, 51, 129-147.
- [27] Khan, Zorina (2004), *The Democratization of Invention: Patents and Copyrights in American Economic Development, 1790-1920*, Cambridge University Press.
- [28] Khandelwal, Amiti (2010), "The long and short (of) quality ladders." *Review of Economic Studies*, 77(4), 1450-1476.
- [29] Kim, Yee Kyoung, Lee, Keun, Park, Walter G., and Choo, Kineung (2012), "Appropriate intellectual property protection and economic growth in countries at different levels of development." *Research Policy*, 41(2), 358-375.
- [30] Liu, Hongkui, Xu, Ming, and Zhong, Tenglong (2019), "Import competition and export markups: evidence from Chinese multi-product exporters." *Annals of Economics and Finance*, 20(1), 357-385.

- [31] Maskus Keith E. and Yang, Lei (2018), “The impacts of post-TRIPS patent reforms on the structure of exports.” *Canadian Journal of Economics*, 51(2), 483-509.
- [32] Merges, Robert and Duffy, John F. (2017), *Patent Law and Policy: Cases and Materials*, 7th edition, Carolina Academic Press, Durham NC.
- [33] Merges, Robert and Nelson, Richard R. (1990), “The complex economics of patent scope,” *Columbia Law Review*, 90(4), 839-916.
- [34] Park, Walter G., (2008), “International patent protection: 1960-2005.” *Research Policy*, 37(4), 761-766.
- [35] Rauch, James E. (1999), “Networks versus markets in international trade.” *Journal of International Economics*, 48(1), 7-35.
- [36] Saggi, Kamal (2016), “Trade, intellectual property rights, and the World Trade Organization.” in K. Bagwell and R. Staiger (eds.), *Handbook of Commercial Policy*, 1B, North-Holland, 433-512.
- [37] Sampson, Thomas (2016), “Dynamic selection: An idea flows theory of entry, trade, and growth.” *Quarterly Journal of Economics*, 131(1), 315-380.
- [38] Schott, Peter K. (2004), “Across-product versus within-product specialization in international trade.” *Quarterly Journal of Economics*, 119(2), 647-678.
- [39] Schott, Peter K. (2008), “The relative sophistication of Chinese exports.” *Economic Policy*, Vol. 53, 5-49.
- [40] Shin, Wonkyu, Lee, Keun, and Park, Walter G. (2016), “When an importer’s protection of IPR interacts with an exporter’s level of technology: comparing the impacts on the exports of the north and south.” *World Economy*, 39(6), 772-802.
- [41] Yang, Guifang & Maskus, Keith E. (2001), “Intellectual property rights and licensing: An econometric investigation. *Review of World Economics*, 137(1), 58-79.

Appendix

I. Countries in the Sample

Income Group	Country	Percent Share of Sample	Count of Products
Upper Middle	Algeria	0.63	23
Upper Middle	Argentina	4.77	174
Upper Middle	Brazil	5.97	218
Upper Middle	Chile	2.80	102
Upper Middle	China, P.R.	5.92	216
Upper Middle	Colombia	3.84	140
Upper Middle	Costa Rica	2.49	91
Upper Middle	Dominican Rep.	0.77	28
Upper Middle	Ecuador	1.12	41
Upper Middle	Iran	0.82	30
Upper Middle	Malaysia	4.82	176
Upper Middle	Mauritius	0.60	22
Upper Middle	Mexico	5.23	191
Upper Middle	Panama	3.75	137
Upper Middle	Peru	2.47	90
Upper Middle	South Africa	5.59	204
Upper Middle	Thailand	4.80	175
Upper Middle	Tunisia	2.60	95
Upper Middle	Turkey	4.91	179
Upper Middle	Uruguay	2.11	77
Upper Middle	Venezuela	2.03	74
Lower Middle	Bolivia	0.30	11
Lower Middle	Cameroon	0.55	20
Lower Middle	Egypt	2.22	81
Lower Middle	El Salvador	1.81	66
Lower Middle	Ghana	0.33	12
Lower Middle	Guatemala	2.77	101
Lower Middle	Guyana	0.19	7
Lower Middle	Honduras	0.99	36
Lower Middle	India	5.67	207
Lower Middle	Indonesia	2.93	107
Lower Middle	Morocco	2.63	96
Lower Middle	Nicaragua	0.38	14
Lower Middle	Paraguay	0.38	14
Lower Middle	Philippines	3.81	139
Lower Middle	Senegal	0.63	23
Lower Middle	Sri Lanka	1.81	66
Lower Middle	Syria	1.04	38
Low	Benin	0.03	1
Low	Kenya	2.27	83
Low	Malawi	0.11	4
Low	Mali	0.11	4
Low	Niger	0.03	1
Low	Sierra Leone	0.05	2
Low	Uganda	0.03	1
Low	Zimbabwe	0.88	32
Total Observations		18,245	

II. Industries, 2-Digit SITC Revision 1

Industry Name	Industry Code	Percent Share of Sample	Count of Exporters
Chemical materials and products, n.e.s	59	2.850	31
Clothing	84	4.740	42
Dyeing, tanning and coloring materials	53	1.04	23
Electrical machinery, apparatus, & appliances	72	8.66	35
Fertilizers, manufactured	56	1.45	27
Footwear	85	0.90	33
Furniture	82	0.96	35
Iron and steel	67	1.07	15
Leather manufactures n.e.s., & dressed fur skins	61	2.71	32
Machinery, other than electrical	71	14.58	38
Manufactures of metals, n.e.s.	69	9.13	32
Medicinal and pharmaceutical products	54	2.06	32
Miscellaneous manufactured articles	89	9.95	36
Non-metallic mineral manufactures, n.e.s.	66	8.14	30
Paper, paperboard and manufactures thereof	64	1.34	29
Perfume materials; toilet and cleansing preparations	55	3.48	33
Rubber manufactures, n.e.s.	62	2.63	32
Sanitary, plumbing, heating, & lightning fixtures	81	1.48	25
Scientific and control instruments, etc.	86	4.14	35
Textile yarn, fabrics, made-up articles, etc.	65	8.99	38
Transport equipment	73	5.04	36
Travel goods, handbags, and similar articles	83	0.82	30
Wood and furniture (excl. manufactures of cork wood)	63	3.84	35
Total Observations		18,245	

III. Data Sources

Variable	Description	Source
Export Quality	Based on unit values adjusted for differences in production cost, pricing, and distance bias	IMF, Henn et al. (2017)
Index of Patent Protection	Based on elements of patentee rights	Park (2008)
Legal Enforcement Effectiveness	Legal security, contract, enforcement, rule of law	Fraser Institute https://www.fraserinstitute.org/
Distance to Frontier	Closeness to World Frontier	Computed from Export Quality (see text)
GDP per capita	Gross Domestic Product per capita	World Bank <i>World Development Indicators</i>
Human Capital	Percentage of completed secondary school attainment in the total population	http://www.barrolee.com/
FDI Net Inflows % of GDP	Foreign Direct Investment inflows as a % of GDP	World Bank <i>World Development Indicators</i>
Import Tariff %	Most Favored Nation applied tariffs (weighted by corresponding trade values)	World Bank <i>World Integrated Trade Solution</i>

Table 1: **Descriptive Statistics**

Variable	Mean	Std.Dev.	Min	Max
Export Quality Index (<i>QLTY</i>)	0.835	0.110	0.0936	1.189
Effective Patent Protection (<i>PRI</i>)	11.90	6.247	0.891	30.15
Distance to Frontier (<i>CTF</i>)	47.48	17.15	0.136	98.92
GDP per capita (<i>GDP_PC</i>)	3033	2027	185.1	7894
Secondary School Attain. (<i>HC</i>)	16.89	9.604	0.500	50.90
FDI Net Inflows, %GDP (<i>FDI</i>)	1.838	1.765	-3.613	11.97
Tariff (<i>TARIFF</i>)	19.97	15.81	0.0100	258.5

Number of Observations = 18,245
(# of obs. with Tariff = 14,145)

Table 2: **Preliminary Check on Stage-Dependence**

	Below Median <i>PRI</i>	Above Median <i>PRI</i>	Mean Difference
Mean Quality Index	0.740	0.730	0.01***
No. of Observations	2915	1646	

(a) **Bottom Quartile (25%) Group of *CTF***

	Below Median <i>PRI</i>	Above Median <i>PRI</i>	Mean Difference
Mean Quality Index	0.900	0.920	-0.02***
No. of Observations	1788	2773	

(b) **Top Quartile (25%) Group of *CTF***

CTF denotes closeness to frontier and *PRI* effective patent protection

Table 3: **Trends in Export Quality and Effective Patent Protection**

Income Group	Variable	Year 1990		Year 2010		
		Obs.	Mean	Obs.	Mean	Mean 2010/ Mean 1990
Low ($CTF^{Norm}=1.00$)	<i>QLTY</i>	828	0.754	128	0.676	0.90
	<i>PRI</i>	828	5.806	128	13.31	2.29
Lower Middle ($CTF^{Norm}=1.18$)	<i>QLTY</i>	2057	0.831	1038	0.806	0.97
	<i>PRI</i>	2057	6.827	1038	16.25	2.38
Upper Middle ($CTF^{Norm}=1.25$)	<i>QLTY</i>	764	0.868	2483	0.875	1.01
	<i>PRI</i>	764	7.906	2483	19.40	2.45

CTF^{Norm} is the closeness-to-frontier measure normalized to the mean value of the low-income group. *QLTY* is the index of export product quality and *PRI* effective patent protection

Table 4: Baseline Results: Patent Protection on Export Quality

Dependent Var.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pooled <i>QLTY</i>	Low Income Countries <i>QLTY</i>	Lower Middle Income Countries <i>QLTY</i>	Upper Middle Income Countries <i>QLTY</i>	Non-High IP Products <i>QLTY</i>	High IP Products <i>QLTY</i>	Low Patent Intensity Products <i>QLTY</i>	High Patent Intensity Products <i>QLTY</i>
<i>PRI</i>	-0.097*** (0.024)	-0.079* (0.041)	-0.116*** (0.030)	-0.086** (0.033)	-0.107*** (0.023)	-0.074*** (0.027)	-0.108*** (0.028)	-0.092*** (0.024)
$PRI \times CTF$	0.027*** (0.007)	0.020 (0.012)	0.032*** (0.007)	0.023** (0.008)	0.030*** (0.006)	0.019*** (0.007)	0.030*** (0.008)	0.025*** (0.006)
<i>GDP_PC</i>	0.085*** (0.016)	0.094** (0.039)	0.047* (0.025)	0.019 (0.020)	0.089*** (0.022)	0.079*** (0.011)	0.088*** (0.020)	0.083*** (0.013)
<i>HC</i>	0.032 (0.055)	0.082 (0.079)	-0.014 (0.053)	-0.047 (0.043)	0.028 (0.073)	0.038 (0.042)	0.021 (0.058)	0.043 (0.056)
<i>FDI</i>	-0.063 (0.108)	1.340** (0.510)	-0.127 (0.102)	0.163 (0.156)	-0.106 (0.142)	0.004 (0.090)	-0.093 (0.108)	-0.032 (0.121)
<i>TARIFF</i>	0.002** (0.001)	0.011* (0.006)	0.006*** (0.002)	-0.000 (0.001)	0.003 (0.002)	0.001 (0.001)	0.001 (0.002)	0.003* (0.002)
<i>Constant</i>	-0.865*** (0.128)	-0.946*** (0.232)	-0.561*** (0.184)	-0.299 (0.178)	-0.933*** (0.177)	-0.780*** (0.090)	-0.895*** (0.162)	-0.842*** (0.109)
Observations N	14,134	1,440	5,938	6,354	7,158	6,976	6,925	7,209
No. singletons	0	277	90	35	0	0	0	0
Original N	14,134	1,717	6,028	6,389	7,158	6,976	6,925	7,209
Adj. R-squared	0.806	0.843	0.798	0.817	0.809	0.801	0.796	0.825
Implied Threshold	36.3	--	37.5	42.1	35.4	49.1	36.6	39.6

Robust standard errors, clustered at the country level, are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.
Country, product, and year fixed effects are included. The classification of high IP products is based on
Delgado et al. (2013). The classification of patent intensity is based on Hu and Png (2012). The implied
threshold is calculated using equation (14). Singletons (one observation per group) are dropped.

Table 5: **Alternative Measures of CTF and PRI**

Dependent Var.	(1) $QLTY$	(2) $QLTY$	(3) $QLTY$
PRI	-0.864*** (0.076)		
$PRI \times CTF^{Alt}$	0.199*** (0.017)		
PRI^{Alt}		-0.195*** (0.105)	-1.337*** (0.583)
$PRI^{Alt} \times CTF$		0.050*** (0.027)	
$PRI^{Alt} \times CTF^{Alt}$			0.306*** (0.074)
GDP_PC	0.093*** (0.018)	0.104*** (0.019)	0.108*** (0.023)
HC	0.043 (0.067)	0.037 (0.075)	0.047 (0.082)
FDI	-0.108 (0.121)	-0.074 (0.126)	-0.116 (0.130)
$TARIFF$	0.002** (0.001)	0.002** (0.001)	0.003*** (0.001)
Observations	14,134	14,134	14,134
Adj. R-squared	0.854	0.797	0.820
Implied Threshold	76.8	49.4	79.0

Robust standard errors, clustered at the country level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Country, product, and year fixed effects are included.

See (14) for the threshold formula.

Table 6: **Robustness Check: Sample Sensitivities**

Dependent Var.	<i>Excluding:</i>	
	Top 5% and Lowest 5% Quality (1) <i>QLTY</i>	China (2) <i>QLTY</i>
<i>PRI</i>	-0.064*** (0.017)	-0.131*** (0.022)
<i>PRI</i> $\times$ <i>CTF</i>	0.018*** (0.005)	0.036*** (0.006)
<i>GDP_PC</i>	0.074*** (0.012)	0.052** (0.024)
<i>HC</i>	0.008 (0.042)	0.054 (0.066)
<i>FDI</i>	0.049 (0.073)	-0.118 (0.106)
<i>TARIFF</i>	0.002** (0.001)	0.002 (0.001)
Observations	12,709	13,258
Adj. R-squared	0.779	0.807
Implied Threshold	35.0	38.0

Robust standard errors, clustered at the country level, are in parentheses.

Country, product, and year fixed effects are included.

*** p<0.01, ** p<0.05, * p<0.1. See (14) for the threshold formula.

Table 7: IV estimation

	Stage 1		Stage 2
	ΔPRI (1)	$\Delta PRI \times CTF$ (2)	$\Delta QLTY$ (3)
Non-colony dummy	0.092*** (0.015)	-0.027 (0.075)	
Non-colony $\times CTF$	0.057*** (0.017)	2.72*** (0.065)	
ΔPRI			0.037 (0.043)
$\Delta PRI \times CTF$			0.010*** (0.003)
ΔGDP_PC	0.621*** (0.064)	3.49*** (0.254)	-0.052** (0.023)
ΔHC	0.018*** (0.001)	0.084*** (0.006)	-0.001* (0.0008)
ΔFDI	0.033*** (0.002)	0.133 (0.010)	-0.001 (0.001)
$\Delta TARIFF$	-0.016** (0.007)	-0.007 (0.028)	0.001 (0.002)
Constant	-0.026 (0.020)	0.045 (0.092)	0.004 (0.005)

No. of observations = 3,524. 2SLS estimator. *** p<0.01, ** p<0.05, * p<0.1.

Changes are measured as differences in natural logarithms.

Robust standard errors are in parentheses, clustered by country.

Underidentification test (Kleibergen-Paap rk LM statistic): $\chi^2=45.5$, p -value=0.00

Weak identification test (Kleibergen-Paap Wald rk F statistic): 23.8

Endogeneity test of endogenous regressors: $\chi^2=3.34$, p -value=0.19

Table 8: **Distribution of *CTF* by Income Group, during 1985-1990**

(1) Range of <i>CTF</i>	(2) Pooled Countries	(3) Upper-Middle Income	(4) Lower-Middle Income	(5) Low Income
<i>Counts of Country-Product Combinations:</i>				
0-10	92	44	27	21
10-20	206	70	88	48
20-30	334	119	184	31
30- 36.3	281	97	173	11
36.3 -50	889	497	379	13
50-60	902	780	120	2
60-70	710	657	52	1
70-80	205	193	11	1
80-90	28	24	4	0
90-100	2	2	0	0
# below threshold	913	481	512	128
# above threshold	2736	2002	526	0
% below threshold	25.0%	19.4%	49.3%	100.0%
% above threshold	75.0%	80.6%	50.7%	0.0%
Total Counts	3649	2483	1038	128

The threshold value **36.3** is for the pooled sample; see Col. 1, Table 4.
The threshold for upper middle income countries is 42.1; see Col. 3, Table 4.
The threshold for lower middle income countries is 37.5; see Col. 4, Table 4.
Based on the product classification in SITC 4-digit, Revision 1. All countries
in our sample are developing countries, as classified by the World Bank.